

**BELIMO**

# Energy Valve Coil Estimator

*(Belimo Energy Valve Coil Estimator.xlsm)*

## USER MANUAL

Release 4.2

Prepared for Belimo internal staff – not for distribution.

Belimo U.S., Inc. | 33 Turner Road, Danbury, CT 06810

Languages: English · Español · Português · Français | Units: US or Metric

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# 1. Introduction

The Belimo Energy Valve Coil Estimator (the “tool”, file name Belimo Energy Valve Coil Estimator.xlsm) is an Excel workbook that answers one question about one air handler: *“So... how much will this smart valve save me on this coil?”* You enter the coil’s design data, how long it runs, where the building is, and the Delta-T the coil delivers today and the Delta-T an Energy Valve will hold. The tool works out how much less water the pump has to move, what that saves in energy and money, and what the investment is worth over 5, 10, 15 and 20 years.

More flow costs money, and it costs it disproportionately: pump power varies with roughly the cube of flow, so cutting flow by a third cuts pump power by around 70%. An Energy Valve holds the coil at the Delta-T you ask for, so the flow – and the pumping bill – comes down.

## Scope of the tool

This is a **coil-level** tool and it counts **pumping savings only**. Its companion, the Energy Valve Savings Estimator, models a whole chilled-water plant and adds the chiller-side savings. Use this tool when the conversation is about a specific coil or air handler; use the plant tool when it is about the central plant. It is an estimating tool for internal use. Results depend on the accuracy of the data entered and do not take external factors into account.

## 1.1 How the workbook is laid out

| Tab          | What it is for                                                                       |
|--------------|--------------------------------------------------------------------------------------|
| Home         | Everything you fill in, in five numbered steps, plus the headline financial results. |
| Pump Savings | The calculation engine: one block per Delta-T, plus Energy Valve model selection.    |
| Flow Curves  | Reference curves relating the control signal to flow through the valve.              |
| Charts       | The same relationships plotted.                                                      |
| Results      | The customer-facing summary. This is the page you present.                           |

## 1.2 What’s in this manual

- Section 2 – Getting started: opening the file, the About screen, updates, which cells you can type in, language and units.
- Section 3 – The Home tab, step by step.
- Section 4 – The Pump Savings tab.
- Section 5 – Flow Curves and Charts.
- Section 6 – The Results tab.

- Section 7 – The financial results explained.
- Section 8 – How the numbers are calculated (for anyone who wants to check the arithmetic).
- Section 9 – Practicalities and troubleshooting. Appendix – Release history.

## 2. Getting Started

### 2.1 Opening the file

Open the workbook in Excel for Windows. It contains macros, so if Excel shows a yellow **Protected View** bar, click **Enable Editing**, and if it then shows a **Security Warning** bar, click **Enable Content**. (For a file that came from e-mail or a download, right-click it in File Explorer, choose Properties, tick **Unblock** and click OK first.)

On opening, the tool quietly checks Belimo's update source for a newer release, takes you to the Home tab, and re-applies your saved language. The workbook opens with a worked example already filled in (a 310 GPM coil in the North East) so you can see every result before you enter your own data. Overwrite the example values in Steps 2 to 5 with your own.

### 2.2 The About screen

Click the small round **(i)** button at the top right of the Home tab (beside the Step 1 callout) to open the About screen. It shows the tool's name, version and release date, a short description, and who maintains the tool.

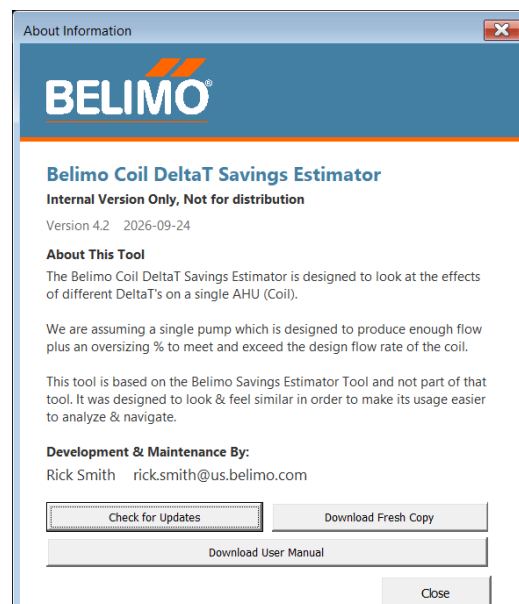


Figure 1 – The About screen.

| Button                      | What it does                                                                                                                           |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Check for Updates</b>    | Asks the update source whether a newer release exists. If one does, you are told what is new and may choose to download it.            |
| <b>Download Fresh Copy</b>  | Downloads the current published release straight away, even if you are already up to date – useful if you think your copy is damaged.  |
| <b>Download User Manual</b> | Saves the current published copy of this manual (a PDF) to a folder you choose. If none has been published yet, the tool tells you so. |

| Button | What it does             |
|--------|--------------------------|
| Close  | Closes the About screen. |

When an update is available you are offered a summary of what changed. If you choose to install it, you pick where to save the new copy through a normal Windows **Save As** dialog. Your current copy is never changed or deleted. If the update source cannot be reached, the check does nothing and the tool works as normal.

## 2.3 The step callouts

The Home tab is laid out as a wizard. Each orange section band has a numbered blue **Step** callout to its right telling you what to enter in that block. Work top to bottom and the tool takes care of the rest.

## 2.4 Which cells can I type in?

| Shading           | Meaning                                                                           |
|-------------------|-----------------------------------------------------------------------------------|
| Pale / near-white | Editable. Type here. (The key “Editable Field” / “Label” is shown on the sheets.) |
| Blue-grey         | A label or a calculated result. Protected – the tool writes it.                   |
| Orange band       | A section heading.                                                                |

The sheets are protected so a stray keystroke cannot break a formula. If a cell will not accept typing, that is the tool telling you it is a result rather than an input.

## 2.5 Language and units

| Control  | Cell | Choices and effect                                                                                                 |
|----------|------|--------------------------------------------------------------------------------------------------------------------|
| Language | F5   | English, Español, Português or Français. Translates every label, heading, unit word and help message on every tab. |
| Units    | F6   | US or Metric. Switches every flow, head, temperature, power and volume figure and its unit.                        |

The calculation always runs in US units internally and converts for display, so switching back and forth never drifts. Figure 2 shows the same Home tab in Spanish with Metric units. Note that the values you have already entered are converted, not re-typed: 310 GPM becomes 1173.478 L/min and a 12 °F Delta-T becomes 6.667 K (shown as °C).

Belimo Energy Valve  
Estimador de Ahorros  $\Delta T$  de Serpentina

Selección de Idioma y Unidades

Idioma

Unidades

Español  
Metric

Datos de Entrada de Diseño de la Serpentina

|                                                                      | Valor    | Unidad  |
|----------------------------------------------------------------------|----------|---------|
| Costo de Electricidad (Promedio)                                     | 0.10     | \$/kWh  |
| Carga de Diseño de la Serpentina                                     | 1173.478 | LPM     |
| Número de Serpentin                                                  | 1        | uds.    |
| Horas de Operación / Día                                             | 20       | horas   |
| Días de Operación / Semana                                           | 6        | días    |
| Semanas de Operación / Año                                           | 52       | semanas |
| Altura Real de la Bomba Primaria (Diseño)                            | 89.672   | kPa     |
| Eficiencia de Bomba, VFD, Motor (Promedio)                           | 70       | %       |
| Delta T de Diseño                                                    | 6.667    | °C      |
| Delta T Real (Antes de la Instalación de Energy Valve)               | 3.333    | °C      |
| Delta T Futuro (Configuración del Gestor de Delta T de Energy Valve) | 6.667    | °C      |

Datos de Entrada de Tiempo de Operación

|                           |      |         |
|---------------------------|------|---------|
| Horas de Operación / Día  | 20   | horas   |
| Días de Operación / Semar | 6    | días    |
| Semanas de Operación / A  | 52   | semanas |
| Horas de Operación Anua   | 5497 | horas   |

Campo Editable

Etiqueta

Paso 1

Selecione su Idioma y sistema de Unidades preferido

Paso 2

Ingrese en los recuadros de la izquierda, lo siguiente:  
Costos Eléctricos  
GPM de Diseño de la Serpentina  
Horas de Operación  
Altura de la Bomba

Paso 3

Ingrese en los recuadros de la izquierda, lo siguiente:

Figure 2 – The top of the Home tab in Spanish with Metric units.

Belimo Energy Valve Coil Estimator – User Manual (Release 4.2)

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## 3. The Home Tab – the Five Steps

Everything you type lives on the Home tab. Figure 3 shows Steps 1 and 2.

### 3.1 Steps 1–2: Language and coil design data

The screenshot shows the 'Belimo Energy Valve Coil ΔT Savings Estimator' interface. It is divided into three main sections: 'Language & Unit Selection', 'Coil Design Input Data', and 'Operating Time Input Data'. On the right side, there are three steps: Step 1 (Language & Unit Selection), Step 2 (Coil Design Input Data), and Step 3 (Operating Time Input Data). Step 1 includes a dropdown for 'Language' (set to English) and 'Units' (set to US). Step 2 includes a table for 'Coil Design Input Data' with columns for 'Field', 'Value', and 'Unit'. Step 3 includes a table for 'Operating Time Input Data' with columns for 'Field', 'Value', and 'Unit'. The 'Coil Design Input Data' table has example values: Electricity Cost (Average) 0.10 \$/kWh, Design Coil Load 310 GPM, Number Of Coils 1 pcs., Operating Hours / Day 20 hours, Operating Days / Week 6 days, Operating Weeks / Year 52 weeks, Actual Primary Pump Head (Design) 30 ft., Pump, VFD, Motor Efficiency (Average) 70 %, Design Delta T 12 °F, Actual Delta T (Prior To Energy Valve Installation) 6 °F, and Future Delta T (Energy Valve Delta T Manager Setting) 12 °F. The 'Operating Time Input Data' table has example values: Operating Hours / Day 20 hours, Operating Days / Week 6 days, Operating Weeks / Year 52 weeks, and Annual Operating Hours 5497 hours. There are also instructions on the right side for each step.

| Field                                                 | Value | Unit   |
|-------------------------------------------------------|-------|--------|
| Electricity Cost (Average)                            | 0.10  | \$/kWh |
| Design Coil Load                                      | 310   | GPM    |
| Number Of Coils                                       | 1     | pcs.   |
| Operating Hours / Day                                 | 20    | hours  |
| Operating Days / Week                                 | 6     | days   |
| Operating Weeks / Year                                | 52    | weeks  |
| Actual Primary Pump Head (Design)                     | 30    | ft.    |
| Pump, VFD, Motor Efficiency (Average)                 | 70    | %      |
| Design Delta T                                        | 12    | °F     |
| Actual Delta T (Prior To Energy Valve Installation)   | 6     | °F     |
| Future Delta T (Energy Valve Delta T Manager Setting) | 12    | °F     |

| Field                  | Value | Unit  |
|------------------------|-------|-------|
| Operating Hours / Day  | 20    | hours |
| Operating Days / Week  | 6     | days  |
| Operating Weeks / Year | 52    | weeks |
| Annual Operating Hours | 5497  | hours |

Figure 3 – Steps 1 and 2 – language and units, and the coil design data (with the example values).

Step 1 sets language and units (Section 2.5). Step 2 is the coil itself – the values every later number is built from.

| Field                             | Cell | Unit   | What it is                                                                                                                                                        |
|-----------------------------------|------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Electricity Cost (Average)        | F10  | \$/kWh | Blended all-in rate. Every dollar figure in the workbook runs through this one number.                                                                            |
| Design Coil Load                  | F11  | GPM    | The coil's design <b>flow</b> . Despite the word "Load", this is a flow rate in gallons per minute (or L/min in Metric), not a tonnage – note the unit beside it. |
| Number Of Coils                   | F12  | pcs.   | Fixed at 1 – this tool estimates one coil at a time. (Its help icon notes multiple coils as a future release.)                                                    |
| Operating Hours / Day             | F13  | hours  | With the two below, sets annual run hours.                                                                                                                        |
| Operating Days / Week             | F14  | days   |                                                                                                                                                                   |
| Operating Weeks / Year            | F15  | weeks  |                                                                                                                                                                   |
| Actual Primary Pump Head (Design) | F16  | ft.    | Head the pump works against at design flow (kPa in Metric).                                                                                                       |

| Field                                                 | Cell | Unit | What it is                                                                               |
|-------------------------------------------------------|------|------|------------------------------------------------------------------------------------------|
| Pump, VFD, Motor Efficiency (Average)                 | F17  | %    | Combined wire-to-water efficiency. If you do not know it, the help icon suggests 70–80%. |
| Design Delta T                                        | F18  | °F   | What the coil was selected for.                                                          |
| Actual Delta T (Prior To Energy Valve Installation)   | F19  | °F   | What it achieves today, before the Energy Valve – the “before” case.                     |
| Future Delta T (Energy Valve Delta T Manager Setting) | F20  | °F   | What the Energy Valve’s Delta-T Manager will hold – the “after” case.                    |

The two Delta-T fields at the bottom are the whole estimate. Actual and Future are the before-and-after the entire comparison is built from, and the tool enforces a minimum gap of 2 °F between them (Future must be at least 2 degrees above Actual) so the pair always makes physical sense; a value outside the allowed range is refused when you type it.

### Delta-Ts snap to six fixed scenarios: 4, 6, 8, 10, 12 and 14 °F

The Results tab compares six fixed Delta-T scenarios: **4, 6, 8, 10, 12 and 14 °F** (in Metric the same six steps, for example 3.33 for 6 °F). Actual and Future must be one of them. If you type another value, the tool adjusts it to the nearest scenario and tells you: 7 becomes 8, 13 becomes 14, and an Actual below 4 becomes 4. Actual can be at most 12 and Future at least 6, and Future is always at least 2 degrees above Actual. The adjusted values are what appear in the box and on the Results tab.

## 3.2 Steps 3–4: Operating time and climate region

Annual Operating Hours

5497

hours

Climate Region

North East

| Point | Bin    | Load | % of Operating Time | Operating Hours |
|-------|--------|------|---------------------|-----------------|
| A     | Design | 100% | 25.0%               | 1374            |
| B     | Peak   | 75%  | 50.0%               | 2748            |
| C     | Low    | 50%  | 25.0%               | 1374            |
| D     | Min    | 25%  | 0.0%                | 0               |
|       |        |      | 100%                | 5497            |

Climate Region Degree Day Input Data

| Climate Region     | Cooling Degree Days | Humidity Adjustment | Multiplier |
|--------------------|---------------------|---------------------|------------|
| North East         | 1034                | 30%                 | 0.88       |
| Central            | 1090                | 30%                 | 0.93       |
| East North Central | 975                 | 30%                 | 0.83       |
| West North Central | 1123                | 15%                 | 0.85       |
| South East         | 2264                | 30%                 | 1.93       |
| South              | 2004                | 30%                 | 1.71       |
| South West         | 2750                | 5%                  | 1.89       |
| Mountain           | 1450                | 5%                  | 1.00       |
| North West         | 720                 | 5%                  | 0.50       |
| West               | 1850                | 5%                  | 1.27       |
| Custom Region      | 800                 | 0%                  | 0.52       |
| Average US         | 1526                | 19%                 | 1.19       |

Selected Value

0.88

Climate region from the drop down box

% of Operating Time  
Enter Percentages of operating time.

Step 4

Verify your Climate Region or select custom and insert your own value.

U.S. Standard Regions

Figure 4 – Steps 3 and 4 – operating time, load points and climate region.

**Annual operating hours** are computed from your three entries and then adjusted for climate: hours per day × days per week × weeks per year × the **climate multiplier** for your region (the Selected Value

beneath the region table). With the example values,  $20 \times 6 \times 52 = 6,240$  hours  $\times 0.88$  (North East) = 5,497 hours.

Rather than simulate every hour of the year, the tool uses four representative operating points and gives each a share of the running time. Only the shares for points A, B and C are editable; point D is calculated as whatever is left. Keep A + B + C at 100% or less: the total cell shows “Error, must = 100%” if they add up to more than 100%.

| Point      | Cell | Load | Default share | Meaning                                                                                                                               |
|------------|------|------|---------------|---------------------------------------------------------------------------------------------------------------------------------------|
| A – Design | E33  | 100% | 25%           | Full-load hours.                                                                                                                      |
| B – Peak   | E34  | 75%  | 50%           | The bulk of a typical year.                                                                                                           |
| C – Low    | E35  | 50%  | 25%           | Light load.                                                                                                                           |
| D – Min    | E36  | 25%  | 0%            | Not editable – calculated as whatever is left after A, B and C (0% when they total 100%). Its hours are included in the annual total. |

**Climate Region** (cell E30) is a drop-down. Each region supplies a cooling-degree-day figure and a humidity adjustment, from which the multiplier is calculated relative to the average US region:

| Region             | Cooling degree days | Humidity adjustment | Multiplier |
|--------------------|---------------------|---------------------|------------|
| North East         | 1034                | 30%                 | 0.88       |
| Central            | 1090                | 30%                 | 0.93       |
| East North Central | 975                 | 30%                 | 0.83       |
| West North Central | 1123                | 15%                 | 0.85       |
| South East         | 2264                | 30%                 | 1.93       |
| South              | 2004                | 30%                 | 1.71       |
| South West         | 2750                | 5%                  | 1.89       |
| Mountain           | 1450                | 5%                  | 1.00       |
| North West         | 720                 | 5%                  | 0.50       |
| West               | 1850                | 5%                  | 1.27       |
| Custom Region      | 800 (editable)      | 0% (editable)       | 0.52       |
| Average US         | 1526                | 19%                 | 1.19       |

For a site that fits none of the standard regions, type your own cooling degree days and humidity adjustment in the **Custom Region** row (cells E53 and F53), then choose “Custom Region” in the drop-down. The multiplier is (region degree days  $\div$  average degree days)  $\times$  (1 + humidity adjustment).

### 3.3 Step 5: Financial inputs and results

Financial Inputs

|                                               |                |                 |                 |                 |
|-----------------------------------------------|----------------|-----------------|-----------------|-----------------|
| Investment Cost (Energy Valve + Installation) | \$4,000.00     | \$              |                 |                 |
| Discount Rate                                 | 4.0%           | %               |                 |                 |
| Estimated Measure Life                        | 15             | years           |                 |                 |
|                                               | <b>5 years</b> | <b>10 years</b> | <b>15 years</b> | <b>20 years</b> |
| Net Present Value (NPV)                       | \$3,488        | \$9,643         | \$14,701        | \$18,859        |
| Savings-Investment Ratio (SIR)                | 1.87           | 3.41            | 4.68            | 5.71            |
| Simple Payback Period                         | 2.4 years      |                 |                 |                 |

Disclaimer

Step 5

Enter the estimated cost to purchase Energy Valves and the cost of installation.

Figure 5 – Step 5 – financial inputs and the three financial results at four horizons.

| Field                                         | Cell | Unit  | What it is                                                                                                                                              |
|-----------------------------------------------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Investment Cost (Energy Valve + Installation) | F62  | \$    | Use the <b>incremental</b> cost over the valve that would otherwise have been fitted, not the whole valve package.                                      |
| Discount Rate                                 | F63  | %     | Converts future dollars into today's. Usually the organisation's cost of capital or required return. 3–6% is common for public and utility-funded work. |
| Estimated Measure Life                        | F64  | years | How long the measure is expected to keep saving. Feeds the lifetime figures on the Results tab.                                                         |

Below the inputs, three results appear at 5, 10, 15 and 20 years: the **Net Present Value (NPV)**, the **Savings-Investment Ratio (SIR)** and the **Simple Payback Period**. All three are explained in Section 7. The small (i) icons beside the Discount Rate, NPV and SIR open a plain-language explanation of each.

### 3.4 The help icons and the Disclaimer

Every round blue (i) icon opens a short help message in your chosen language. On the Home tab they explain Number of Coils, Pump Efficiency, Discount Rate, NPV and SIR; other tabs have their own (Sections 4 and 6). Click Close to dismiss.

A **Disclaimer** band at the foot of the Home tab states that the tool is not copyrighted by Belimo Aircontrols (USA), Inc., is strictly an internal estimating tool, that results depend on the accuracy of the data entered and do not consider external factors, and that it looks at the effects of pump costs over various Delta-Ts for a single coil using standard pump affinity rules.

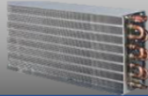
## 4. The Pump Savings Tab

The engine. Nothing here needs filling in except one factor – but it is worth understanding what it is doing.



### Belimo Energy Valve

### Coil ΔT Savings Estimator



Coil Design Data

|                                        |           |                    |             |                                      |
|----------------------------------------|-----------|--------------------|-------------|--------------------------------------|
| Pump Design Factor                     | 125%      | Max Flow Rate      | 387.5 GPM   | Editable Field                       |
| Minimum Pump head                      | 0         |                    |             | Label                                |
| Design Chilled Water Coil Flow [V'max] | 310       |                    |             |                                      |
| Design Chilled Water Coil ΔT           | 12        |                    |             |                                      |
| Design Chilled Water Coil [tons]       | 155       |                    |             |                                      |
| Design Chilled Water Coil [BTU/h]      | 1,860,000 | Energy Valve Model | EV400SU-317 | 95.317 gpm 4" DN100 * based on V'max |

Coil Design Performance Data

| Load [%] | Time [%] | Time [h] | Flow [GPM] | Head [ft.] | Power [HP] | BTU/h     | Energy [kWh] | Annual gallons | Pump Costs | Pump costs     |
|----------|----------|----------|------------|------------|------------|-----------|--------------|----------------|------------|----------------|
| 100      | 25.0     | 1374     | 310        | 30         | 3.4        | 1,860,000 | 3,439        | 25,559,179     | \$344      | Annual gallons |
| 75       | 50.0     | 2748     | 233        | 17         | 1.4        | 1,395,000 | 2,902        | 38,338,769     | \$290      | Avg GPM        |
| 50       | 25.0     | 1374     | 155        | 8          | 0.4        | 930,000   | 430          | 12,779,590     | \$43       | Energy [kWh]   |
| 25       | 0.0      | 0        | 78         | 2          | 0.1        | 465,000   | 0            | 0              | \$0        |                |
|          |          |          |            |            |            |           |              | 6,771          | 76,677,537 | 677            |

Chilled Water Coil Flows at different Delta T [°F]

Chilled Water Coil Flow at Delta T [°F]: 4

| Load [%] | Time [%] | Time [h] | Flow [GPM] | Head [ft.] | Power [HP] | BTU/h   | Energy [kWh] | Annual gallons |
|----------|----------|----------|------------|------------|------------|---------|--------------|----------------|
| 100      | 25.0     | 1374     | 388        | 47         | 6.6        | 775,000 | 6,717        | 31,948,974     |
| 75       | 50.0     | 2748     | 388        | 47         | 6.6        | 775,000 | 13,435       | 63,897,948     |
| 50       | 25.0     | 1374     | 388        | 47         | 6.6        | 775,000 | 6,717        | 31,948,974     |
| 25       | 0.0      | 0        | 233        | 17         | 1.4        | 465,000 | 0            | 0              |
|          |          |          |            |            |            | 26,869  | 127,795,895  |                |

Chilled Water Coil Flow at Delta T [°F]: 6

| Load [%] | Time [%] | Time [h] | Flow [GPM] | Head [ft.] | Power [HP] | BTU/h   | Energy [kWh] | Annual gallons |
|----------|----------|----------|------------|------------|------------|---------|--------------|----------------|
| 100      | 25.0     | 1374     | 388        | 47         | 6.6        | 775,000 | 6,717        | 31,948,974     |
| 75       | 50.0     | 2748     | 388        | 47         | 6.6        | 775,000 | 13,435       | 63,897,948     |
| 50       | 25.0     | 1374     | 388        | 47         | 6.6        | 775,000 | 6,717        | 31,948,974     |
| 25       | 0.0      | 0        | 233        | 17         | 1.4        | 465,000 | 0            | 0              |
|          |          |          |            |            |            | 26,869  | 127,795,895  |                |

| BTU/h          | Increase |
|----------------|----------|
| As % of Design | In GPM   |
| 42%            | 78       |
| 56%            | 155      |
| 83%            | 233      |
| 100%           | 155      |

Figure 6 – The Pump Savings tab: design data, valve selection, the design-point performance table and the first Delta-T block.

### 4.1 Design data

| Field                                      | Cell    | What it is                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pump Design Factor                         | F5      | The only editable cell on this tab (default 125%). A safety margin on design flow – how much bigger the pump is than the coil strictly needs. It caps how much flow the pump can deliver (the <b>Max Flow Rate</b> beside it is design flow × this factor). Leave it alone unless you know the real oversizing. |
| Minimum Pump head                          | F6      | Reserved for a future release; leave at 0.                                                                                                                                                                                                                                                                      |
| Design Chilled Water Coil Flow [V'max]     | F7      | Your Design Coil Load from the Home tab.                                                                                                                                                                                                                                                                        |
| Design Chilled Water Coil ΔT               | F8      | Your Design Delta T.                                                                                                                                                                                                                                                                                            |
| Design Chilled Water Coil [tons] / [BTU/h] | F9, F10 | The design cooling capacity implied by the flow and Delta-T (tons = GPM × ΔT ÷ 24; BTU/h = GPM × ΔT × 500).                                                                                                                                                                                                     |

## 4.2 Energy Valve selection

Using the design flow, the tool picks the Energy Valve model that covers it and shows the model, its flow range and its pipe size beside the design data (for example EV400SU-317, 95–317 gpm, 4" DN100). The choice is based on V<sub>max</sub>, the design flow. The range is shown in whichever units you selected.

| Design flow (GPM) from | Model       | Flow range     | Pipe size  |
|------------------------|-------------|----------------|------------|
| 1.65                   | EV050       | 1.65–6.6 gpm   | 0.5" DN15  |
| 6.6                    | EV075       | 2.7–11 gpm     | 0.75" DN20 |
| 12                     | EV100       | 4.5–18.2 gpm   | 1" DN25    |
| 19                     | EV125       | 7.1–28.5 gpm   | 1.25" DN32 |
| 29                     | EV150       | 11–44 gpm      | 1.5" DN40  |
| 45                     | EV200       | 16.5–66 gpm    | 2" DN50    |
| 67                     | EV200H      | 25–100 gpm     | 2" DN50    |
| 101                    | EV250SU-127 | 38–127 gpm     | 2.5" DN65  |
| 128                    | EV300SU-180 | 54–180 gpm     | 3" DN80    |
| 181                    | EV400SU-317 | 95–317 gpm     | 4" DN100   |
| 318                    | EV500SU-495 | 149–495 gpm    | 5" DN125   |
| 496                    | EV600SU-713 | 214–713 gpm    | 6" DN150   |
| 714                    | Dual EV's   | Up to 1400 gpm | –          |

The selection is a guide for the conversation, not a sizing calculation; confirm the valve with a full sizing before quoting.

## 4.3 The performance blocks

Below the design data, the tab has a block for the design case and then one block for each Delta-T scenario (4, 6, 8, 10, 12 and 14 °F). Each block carries the four load points across flow, head, power, energy and cost, and they all work the same way: for each load point the tool works out the flow needed at that Delta-T, the head the pump sees at that flow, the resulting pump power and, from the hours in that bin, the energy, water volume and cost. Add the four rows and you have that Delta-T's annual pumping bill. Section 8 gives the formulas.

## 4.4 Saving the tab as a PDF

The picture button at the top right of the tab saves the sheet you are on as a PDF. It offers a file name that includes the sheet name and the date and time, in the workbook's folder, through a Save As dialog, and confirms with the file location when done.

#### Help icons on this tab

The (i) beside Pump Design Factor explains it as the percentage of additional pumping available due to over-sizing or future design. The (i) beside Minimum Pump head is marked as a future feature.

## 5. Flow Curves and Charts

Reference material. Nothing to fill in on either tab. These tabs are there for the technical conversation – how a control signal maps to valve position and flow, and what that looks like plotted. They are useful when a customer or controls contractor asks how the valve behaves across its range; they are **not** part of the savings calculation.

### 5.1 Flow Curves

Flow Curves relates the DDC control signal to the actual flow through the valve, at the design flow you entered. Down the left, each row is one step of valve opening (0–100%) with the corresponding control voltage; the columns show the flow for a linear characteristic and for an equal-percentage characteristic, and the heat transfer (BTU/h) at the design Delta-T. The blue columns are the ones to use, and a worked example is printed at the right: an Energy Valve with a V'max of 310 GPM at 2.88 volts flows 17.67 GPM at 6% valve opening, producing 106,023 BTU/h at a 12 °F Delta-T (this example text is fixed and does not change with your inputs). The right-hand block feeds the charts.

| BELIMO® Belimo Energy Valve      |         |                                            |            |        |       |       |         |  |  |
|----------------------------------|---------|--------------------------------------------|------------|--------|-------|-------|---------|--|--|
| Used to Calculate Projected Flow |         |                                            |            |        |       |       |         |  |  |
| DDC Control signal               |         | Valve Position & Flow based on DDC Voltage |            |        |       |       |         |  |  |
| Voltage                          | Voltage | Linear Part                                | Equal Part | V'max  | V'max | ΔT    | btu/h   |  |  |
| 0.50                             | 2.00    | Linear                                     | EQ %       | Linear | EQ %  |       |         |  |  |
| In %                             | In V    | In V                                       | In %       | In %   | In %  |       |         |  |  |
| 0                                | 0.50    | 2.00                                       | 0.0%       | 4.0%   | 0.00  | 0.00  | 0       |  |  |
| 1                                | 0.60    | 2.08                                       | 0.4%       | 4.1%   | 0.00  | 0.00  | 0       |  |  |
| 2                                | 0.69    | 2.16                                       | 0.7%       | 4.3%   | 0.00  | 0.00  | 0       |  |  |
| 3                                | 0.79    | 2.24                                       | 1.1%       | 4.4%   | 0.00  | 0.00  | 0       |  |  |
| 4                                | 0.88    | 2.32                                       | 1.4%       | 4.5%   | 0.00  | 0.00  | 0       |  |  |
| 5                                | 0.98    | 2.40                                       | 1.8%       | 4.7%   | 0.00  | 0.00  | 0       |  |  |
| 6                                | 1.07    | 2.48                                       | 2.1%       | 4.9%   | 0.00  | 0.00  | 0       |  |  |
| 7                                | 1.17    | 2.56                                       | 2.5%       | 5.0%   | 0.00  | 15.54 | 93,210  |  |  |
| 8                                | 1.26    | 2.64                                       | 2.8%       | 5.2%   | 8.68  | 16.04 | 96,260  |  |  |
| 9                                | 1.36    | 2.72                                       | 3.2%       | 5.3%   | 9.77  | 16.57 | 99,410  |  |  |
| 10                               | 1.45    | 2.80                                       | 3.5%       | 5.5%   | 10.85 | 17.11 | 102,663 |  |  |
| 11                               | 1.55    | 2.88                                       | 3.9%       | 5.7%   | 11.94 | 17.67 | 106,023 |  |  |
| 12                               | 1.64    | 2.96                                       | 4.2%       | 5.9%   | 13.02 | 18.25 | 109,492 |  |  |
| 13                               | 1.74    | 3.04                                       | 4.6%       | 6.1%   | 14.11 | 18.85 | 113,076 |  |  |
| 14                               | 1.83    | 3.12                                       | 4.9%       | 6.3%   | 15.19 | 19.46 | 116,776 |  |  |
| 15                               | 1.93    | 3.20                                       | 5.3%       | 6.5%   | 16.28 | 20.10 | 120,597 |  |  |
| 16                               | 2.02    | 3.28                                       | 5.6%       | 6.7%   | 17.36 | 20.76 | 124,544 |  |  |
| 17                               | 2.12    | 3.36                                       | 6.0%       | 6.9%   | 18.45 | 21.44 | 128,619 |  |  |
| 18                               | 2.21    | 3.44                                       | 6.3%       | 7.1%   | 19.53 | 22.14 | 132,828 |  |  |
| 19                               | 2.31    | 3.52                                       | 6.7%       | 7.4%   | 20.62 | 22.86 | 137,175 |  |  |
| 20                               | 2.40    | 3.60                                       | 7.0%       | 7.6%   | 21.70 | 23.61 | 141,664 |  |  |
| 21                               | 2.50    | 3.68                                       | 7.4%       | 7.9%   | 22.79 | 24.38 | 146,300 |  |  |
| 22                               | 2.59    | 3.76                                       | 7.7%       | 8.1%   | 23.87 | 25.18 | 151,087 |  |  |
| 23                               | 2.69    | 3.84                                       | 8.1%       | 8.4%   | 24.96 | 26.01 | 156,031 |  |  |
| 24                               | 2.78    | 3.92                                       | 8.4%       | 8.7%   | 26.04 | 26.86 | 161,137 |  |  |
| 25                               | 2.88    | 4.00                                       | 8.8%       | 8.9%   | 27.13 | 27.74 | 166,410 |  |  |
| 26                               | 2.97    | 4.08                                       | 9.1%       | 9.2%   | 28.21 | 28.64 | 171,856 |  |  |
| 27                               | 3.07    | 4.16                                       | 9.5%       | 9.5%   | 29.30 | 29.58 | 177,480 |  |  |
| 28                               | 3.16    | 4.24                                       | 9.8%       | 9.9%   | 30.38 | 30.55 | 183,287 |  |  |
| 29                               | 3.26    | 4.32                                       | 10.2%      | 10.2%  | 31.47 | 31.55 | 189,285 |  |  |
| 30                               | 3.35    | 4.40                                       | 10.5%      | 10.5%  | 32.55 | 32.58 | 195,480 |  |  |
| 31                               | 3.45    | 4.48                                       | 10.9%      | 10.9%  | 33.64 | 33.65 | 201,876 |  |  |
| 32                               | 3.54    | 4.56                                       | 11.2%      | 11.2%  | 34.72 | 34.75 | 208,483 |  |  |
| 33                               | 3.64    | 4.64                                       | 11.6%      | 11.6%  | 35.81 | 35.88 | 215,305 |  |  |
| 34                               | 3.73    | 4.72                                       | 11.9%      | 12.0%  | 36.89 | 37.06 | 222,351 |  |  |
| 35                               | 3.83    | 4.80                                       | 12.3%      | 12.3%  | 37.98 | 38.27 | 229,627 |  |  |
| 36                               | 3.92    | 4.88                                       | 12.6%      | 12.7%  | 39.06 | 39.52 | 237,141 |  |  |
| 37                               | 4.02    | 4.96                                       | 13.0%      | 13.2%  | 40.15 | 40.82 | 244,901 |  |  |
| 38                               | 4.11    | 5.04                                       | 13.3%      | 13.6%  | 41.23 | 42.15 | 252,916 |  |  |
| 39                               | 4.21    | 5.12                                       | 13.7%      | 14.0%  | 42.32 | 43.53 | 261,192 |  |  |
| 40                               | 4.30    | 5.20                                       | 14.0%      | 14.5%  | 43.40 | 44.96 | 269,739 |  |  |
| 41                               | 4.40    | 5.28                                       | 14.4%      | 15.0%  | 44.49 | 46.43 | 278,566 |  |  |
| 42                               | 4.49    | 5.36                                       | 14.7%      | 15.5%  | 45.57 | 47.95 | 287,682 |  |  |
| 43                               | 4.59    | 5.44                                       | 15.1%      | 16.0%  | 46.66 | 49.52 | 297,096 |  |  |
| 44                               | 4.68    | 5.52                                       | 15.4%      | 16.5%  | 47.74 | 51.14 | 306,818 |  |  |
| 45                               | 4.78    | 5.60                                       | 15.8%      | 17.0%  | 48.83 | 52.81 | 316,859 |  |  |
| 46                               | 4.87    | 5.68                                       | 16.1%      | 17.6%  | 49.91 | 54.54 | 327,228 |  |  |
| 47                               | 4.97    | 5.76                                       | 16.5%      | 18.2%  | 51.00 | 56.32 | 337,936 |  |  |
| 48                               | 5.06    | 5.84                                       | 16.8%      | 18.8%  | 52.08 | 58.17 | 348,994 |  |  |
| 49                               | 5.16    | 5.92                                       | 17.2%      | 19.4%  | 53.17 | 60.07 | 360,415 |  |  |
| 50                               | 5.25    | 6.00                                       | 17.5%      | 20.0%  | 54.25 | 62.03 | 372,209 |  |  |

| Used for Charts |       |      |       |         |  |  |  |  |  |
|-----------------|-------|------|-------|---------|--|--|--|--|--|
| Equal Part      | V'max | Y    | V'max | btu/h   |  |  |  |  |  |
| EQ %            | EQ %  | 2.00 | EQ %  |         |  |  |  |  |  |
| In %            | In V  | In V | In %  | btu/h   |  |  |  |  |  |
| 4.0%            | 0.00  | 2.00 | 0.00  | 0       |  |  |  |  |  |
| 4.1%            | 0.00  | 2.08 | 0.00  | 0       |  |  |  |  |  |
| 4.3%            | 0.00  | 2.16 | 0.00  | 0       |  |  |  |  |  |
| 4.4%            | 0.00  | 2.24 | 0.00  | 0       |  |  |  |  |  |
| 4.5%            | 0.00  | 2.32 | 0.00  | 0       |  |  |  |  |  |
| 4.7%            | 0.00  | 2.40 | 0.00  | 0       |  |  |  |  |  |
| 4.9%            | 0.00  | 2.48 | 0.00  | 0       |  |  |  |  |  |
| 5.0%            | 15.54 | 2.56 | 15.54 | 93,210  |  |  |  |  |  |
| 5.2%            | 16.04 | 2.64 | 16.04 | 96,260  |  |  |  |  |  |
| 5.3%            | 16.57 | 2.72 | 16.57 | 99,410  |  |  |  |  |  |
| 5.5%            | 17.11 | 2.80 | 17.11 | 102,663 |  |  |  |  |  |
| 5.7%            | 17.67 | 2.88 | 17.67 | 106,023 |  |  |  |  |  |
| 5.9%            | 18.25 | 2.96 | 18.25 | 109,492 |  |  |  |  |  |
| 6.1%            | 18.85 | 3.04 | 18.85 | 113,076 |  |  |  |  |  |
| 6.3%            | 19.46 | 3.12 | 19.46 | 116,776 |  |  |  |  |  |
| 6.5%            | 20.10 | 3.20 | 20.10 | 120,597 |  |  |  |  |  |
| 6.7%            | 20.76 | 3.28 | 20.76 | 124,544 |  |  |  |  |  |
| 6.9%            | 21.44 | 3.36 | 21.44 | 128,619 |  |  |  |  |  |
| 7.1%            | 22.14 | 3.44 | 22.14 | 132,828 |  |  |  |  |  |
| 7.4%            | 22.86 | 3.52 | 22.86 | 137,175 |  |  |  |  |  |
| 7.6%            | 23.61 | 3.60 | 23.61 | 141,664 |  |  |  |  |  |
| 7.9%            | 24.38 | 3.68 | 24.38 | 146,300 |  |  |  |  |  |
| 8.1%            | 25.18 | 3.76 | 25.18 | 151,087 |  |  |  |  |  |
| 8.4%            | 26.01 | 3.84 | 26.01 | 156,031 |  |  |  |  |  |
| 8.7%            | 26.86 | 3.92 | 26.86 | 161,137 |  |  |  |  |  |
| 8.9%            | 27.74 | 4.00 | 27.74 | 166,410 |  |  |  |  |  |
| 9.2%            | 28.64 | 4.08 | 28.64 | 171,856 |  |  |  |  |  |
| 9.5%            | 29.58 | 4.16 | 29.58 | 177,480 |  |  |  |  |  |
| 9.9%            | 30.55 | 4.24 | 30.55 | 183,287 |  |  |  |  |  |
| 10.2%           | 31.55 | 4.32 | 31.55 | 189,285 |  |  |  |  |  |
| 10.5%           | 32.58 | 4.40 | 32.58 | 195,480 |  |  |  |  |  |
| 10.9%           | 33.65 | 4.48 | 33.65 | 201,876 |  |  |  |  |  |
| 11.2%           | 34.75 | 4.56 | 34.75 | 208,483 |  |  |  |  |  |
| 11.6%           | 35.88 | 4.64 | 35.88 | 215,305 |  |  |  |  |  |
| 12.0%           | 37.06 | 4.72 | 37.06 | 222,351 |  |  |  |  |  |
| 12.3%           | 38.27 | 4.80 | 38.27 | 229,627 |  |  |  |  |  |
| 12.7%           | 39.52 | 4.88 | 39.52 | 237,141 |  |  |  |  |  |
| 13.2%           | 40.82 | 4.96 | 40.82 | 244,901 |  |  |  |  |  |
| 13.6%           | 42.15 | 5.04 | 42.15 | 252,916 |  |  |  |  |  |
| 14.0%           | 43.53 | 5.12 | 43.53 | 261,192 |  |  |  |  |  |
| 14.5%           | 44.96 | 5.20 | 44.96 | 269,739 |  |  |  |  |  |
| 15.0%           | 46.43 | 5.28 | 46.43 | 278,566 |  |  |  |  |  |
| 15.5%           | 47.95 | 5.36 | 47.95 | 287,682 |  |  |  |  |  |
| 16.0%           | 49.52 | 5.44 | 49.52 | 297,096 |  |  |  |  |  |
| 16.5%           | 51.14 | 5.52 | 51.14 | 306,818 |  |  |  |  |  |
| 17.0%           | 52.81 | 5.60 | 52.81 | 316,859 |  |  |  |  |  |
| 17.6%           | 54.54 | 5.68 | 54.54 | 327,228 |  |  |  |  |  |
| 18.2%           | 56.32 | 5.76 | 56.32 | 337,936 |  |  |  |  |  |
| 18.8%           | 58.17 | 5.84 | 58.17 | 348,994 |  |  |  |  |  |
| 19.4%           | 60.07 | 5.92 | 60.07 | 360,415 |  |  |  |  |  |
| 20.0%           | 62.03 | 6.00 | 62.03 | 372,209 |  |  |  |  |  |

**Projected Flow Calculations**

Example:  
 An Energy Valve with a V'max of 310 GPM  
 At 2.88 volts, you would be flowing 17.67 GPM  
 At 6% Valve Opening  
 Producing 106,023 btu's at a 12 DeltaT

Hint: Refer to the Blue Columns

Figure 7 – The top of the Flow Curves tab.

### 5.2 Charts

The same relationships plotted: percent of valve open to GPM, DDC voltage to GPM, DDC voltage to GPM and BTU/h together, and GPM to BTU/h (a straight line, because heat transfer is flow × Delta-T × 500 at a fixed Delta-T).

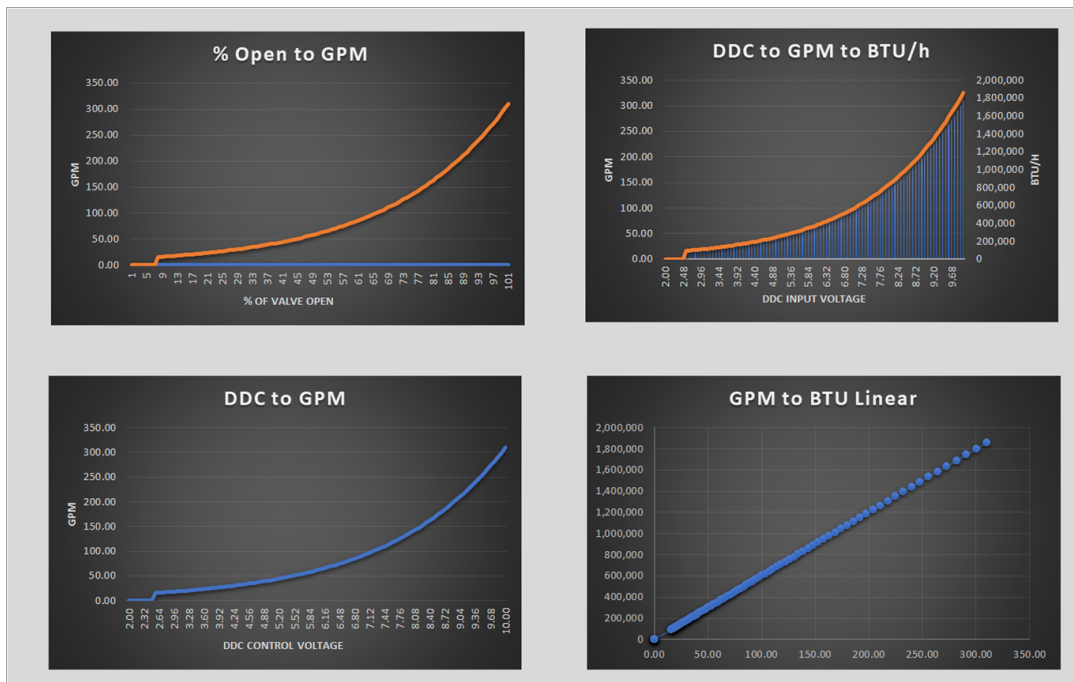


Figure 8 – The Charts tab.

## 6. The Results Tab

The customer-facing page. Nothing to fill in – this is what you present or print. Before and after, the full Delta-T range colour-graded from worst to best, and an interactive slider beneath.

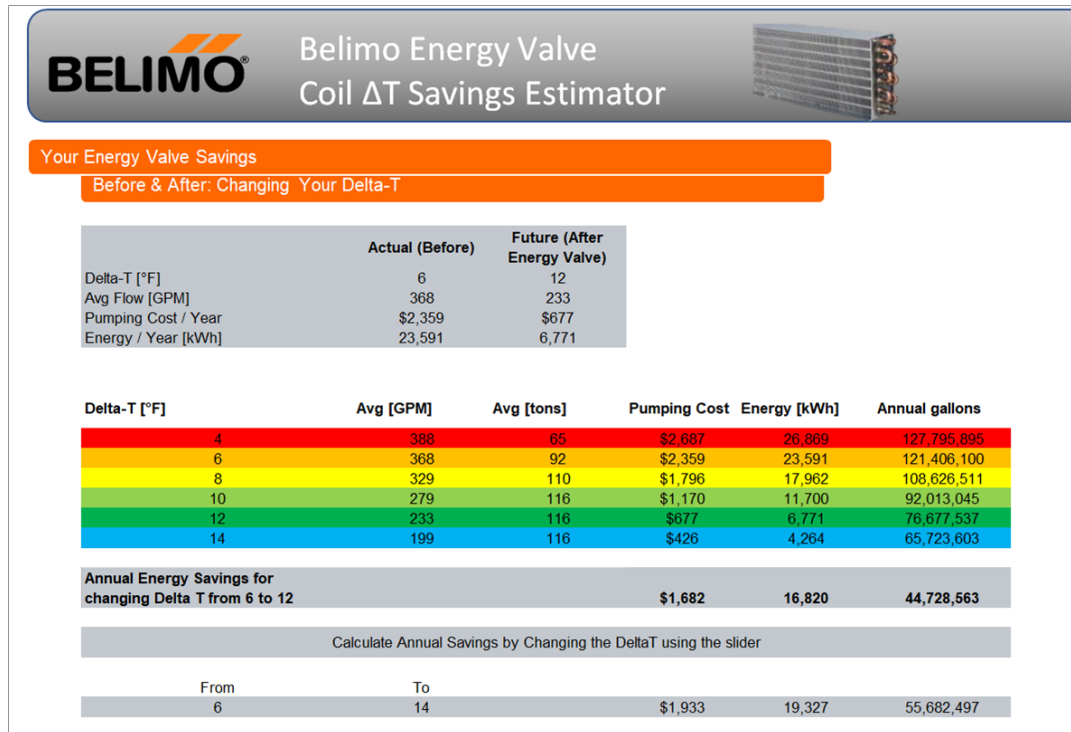


Figure 9 – The top of the Results tab: before and after, the Delta-T scenarios and the slider.

### 6.1 What each block shows

| Block                        | Contents                                                                                                                                                                          |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Before &amp; After</b>    | Your Actual and Future Delta-T side by side: Delta-T, average flow, pumping cost per year and energy per year.                                                                    |
| <b>Delta-T table</b>         | Six scenarios (4–14 °F) colour-graded red (worst) to blue (best), with average flow, average tonnage, pumping cost, energy and annual water volume.                               |
| <b>Annual Energy Savings</b> | The headline line: dollars, kilowatt-hours and gallons saved by moving from your Actual Delta-T to your Future one.                                                               |
| <b>The slider</b>            | Set any “To” Delta-T with the spin buttons and the savings line beneath recalculates live, moving 2 degrees at a time. The “From” case stays pinned to the coil’s Actual Delta-T. |

The slider is the most persuasive thing on the tab: it answers “what if we only get halfway there?” while the customer watches, instead of forcing a single optimistic number. Its (i) icon explains that the slider changes the Delta-T savings by 2 degrees at a time.

### 6.2 Greenhouse gas equivalents and financial summary

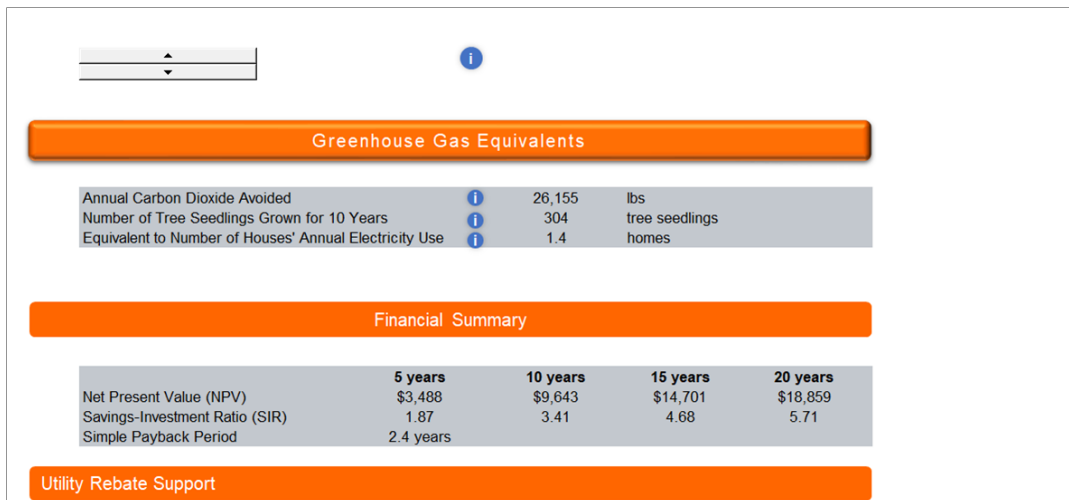


Figure 10 – The slider control, greenhouse gas equivalents and financial summary.

The greenhouse-gas block expresses the saving as pounds of carbon dioxide avoided and as two equivalents people can picture – tree seedlings grown for ten years, and a number of homes' annual electricity use. The (i) icons note that the calculations follow EPA methods; the homes figure is the energy saved divided by 12,258 kWh, the annual electricity use of a home used by the tool. The financial summary repeats the three results from the Home tab at 5, 10, 15 and 20 years.

### 6.3 Utility rebate support

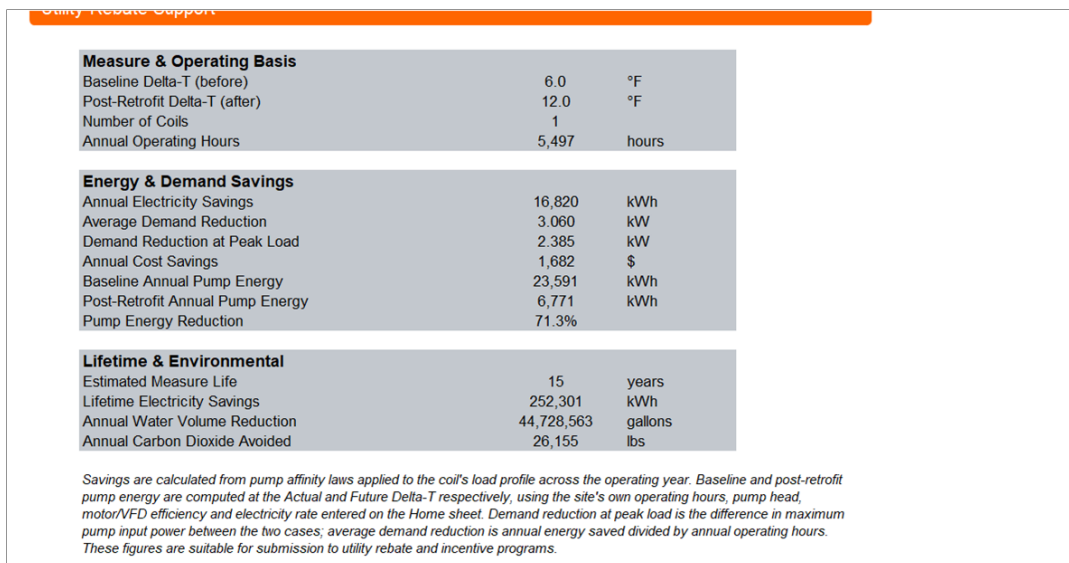


Figure 11 – The Utility Rebate Support block.

This block restates the result in the shape a utility programme wants: the measure basis (baseline and post-retrofit Delta-T, coil count, annual hours), the energy and demand savings, and the lifetime and environmental figures. It closes with a methodology note, so a reviewer can see the calculation basis without having to ask.

### Two demand figures, and they are not the same thing

**Average Demand Reduction** is the annual energy saving divided by annual hours. **Demand Reduction at Peak Load** is the difference in pump power at the design point. A rebate form usually wants the peak figure; do not quote one where the other is asked for.

## 7. The Financial Results

Three numbers, shown at four horizons. These are the ones that get misread, so they are worth a careful read.

### 7.1 First, the idea behind all three: present value

Money in the future is worth less than money now – you could have invested today's dollar in the meantime. "Present value" is what a future amount is worth today, once you have discounted it back at the rate you set in Step 5. At a 4% discount rate, a dollar saved ten years from now counts as about 68 cents today. The discount rate is a lever: raise it and future savings count for less, so both Net Present Value and the Savings-Investment Ratio get smaller. It is not a fudge factor – set it to what your organisation actually requires and leave it there.

### 7.2 Net Present Value (NPV)

Take every year's saving, discount it back to today, add them up, and subtract what you spent on day one. What is left is the value the project adds, expressed in today's dollars. **Above zero** means worth doing; **below zero** means not at this discount rate – which does not mean the project loses money, only that it does not clear the return you demanded of it. It is shown at 5, 10, 15 and 20 years and grows with the horizon, because the spend happens once at the start while the savings keep accruing.

### 7.3 Savings-Investment Ratio (SIR)

The present value of everything you get back, divided by what you put in. Because it is a ratio rather than an amount, it lets you compare projects of very different sizes on equal terms. **Above 1.0** returns more than it costs; **below 1.0** does not pay back in present-value terms. An SIR of 1.9 means every dollar invested returns about \$1.90 in today's money. Many public-sector and utility incentive programmes set a minimum ratio as an eligibility condition, which is why it sits beside NPV on both the Home and Results tabs.

### 7.4 Simple Payback Period

How long the savings take to repay the investment, ignoring the time value of money entirely: the investment divided by the annual cost saving, shown in years. It is the number a building owner asks for first, and the easiest to explain. Read it as a sanity check, not as the whole case – it ignores everything that happens after payback, so it systematically undersells a long-lived measure. Quote it because people ask for it, then show the Net Present Value.

#### Which one do you lead with?

For a building owner, Simple Payback then Net Present Value. For a finance department, Net Present Value. For a utility or public-sector programme, the Savings-Investment Ratio – it is often written straight into the eligibility rules.

## 7.5 Reading the four horizons

Each result is repeated at 5, 10, 15 and 20 years, answering the same question over a different holding period: if we judged this on an N-year view, how would it look? Short horizons always look worse, because the money goes out on day one while the savings accumulate. Quote the horizon that matches how the customer actually evaluates capital – and say which horizon you are quoting.

## 7.6 Worked example (the values the workbook opens with)

| Item                                     | Value                                                                                        |
|------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Coil</b>                              | 310 GPM, 12 °F design Delta-T (155 tons), 30 ft head, 70% efficiency, electricity \$0.10/kWh |
| <b>Operating time</b>                    | 20 h/day, 6 days/week, 52 weeks/year, North East (multiplier 0.88) = 5,497 hours per year    |
| <b>Actual → Future Delta-T</b>           | 6 °F → 12 °F                                                                                 |
| <b>Pumping cost before / after</b>       | \$2,359 per year → \$677 per year (23,591 kWh → 6,771 kWh, a 71.3% reduction)                |
| <b>Annual saving</b>                     | \$1,682 and 16,820 kWh, and 44,728,563 gallons less water pumped                             |
| <b>Investment / discount rate / life</b> | \$4,000 / 4% / 15 years                                                                      |
| <b>NPV at 5 / 10 / 15 / 20 years</b>     | \$3,488 / \$9,643 / \$14,701 / \$18,859                                                      |
| <b>SIR at 5 / 10 / 15 / 20 years</b>     | 1.87 / 3.41 / 4.68 / 5.71                                                                    |
| <b>Simple payback</b>                    | 2.4 years                                                                                    |

## 8. How the Numbers Are Calculated

For anyone who wants to check the arithmetic. All calculations are done in US units and converted for display.

| Quantity                                       | Calculation                                                                                                                                                                  |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Design tonnage</b>                          | Design flow (GPM) × Design Delta-T ÷ 24                                                                                                                                      |
| <b>Flow needed at a load point and Delta-T</b> | Design tonnage × 24 × (load %) ÷ Delta-T, limited to the pump's maximum flow (design flow × Pump Design Factor). In the design case the flow is simply load % × design flow. |
| <b>Pump head at that flow</b>                  | Design head × (flow ÷ design flow) <sup>2</sup> (the pump affinity law; never below Minimum Pump head)                                                                       |
| <b>Pump power (HP)</b>                         | Flow (GPM) × head (ft) ÷ (3960 × efficiency)                                                                                                                                 |
| <b>Energy (kWh)</b>                            | Hours in the bin × HP × 0.746                                                                                                                                                |
| <b>Cost (\$)</b>                               | kWh × electricity cost                                                                                                                                                       |
| <b>Annual water volume (gal)</b>               | Flow × 60 × hours in the bin                                                                                                                                                 |
| <b>Annual operating hours</b>                  | Hours/day × days/week × weeks/year × climate multiplier                                                                                                                      |
| <b>Climate multiplier</b>                      | (Region degree days ÷ average degree days) × (1 + region humidity adjustment)                                                                                                |
| <b>Annual saving</b>                           | Annual pumping cost (or kWh, or gallons) at Actual Delta-T minus the same at Future Delta-T                                                                                  |
| <b>Average demand reduction (kW)</b>           | Annual kWh saved ÷ annual operating hours                                                                                                                                    |
| <b>NPV (N years)</b>                           | – Investment + annual cost saving × $[1 - (1 + r)^{-N}] ÷ r$ (r = discount rate; N × saving if r = 0)                                                                        |
| <b>SIR (N years)</b>                           | Annual cost saving × $[1 - (1 + r)^{-N}] ÷ r ÷$ Investment                                                                                                                   |
| <b>Simple payback</b>                          | Investment ÷ annual cost saving                                                                                                                                              |
| <b>CO<sub>2</sub> avoided (lb)</b>             | kWh saved × 1.555                                                                                                                                                            |
| <b>Tree seedlings</b>                          | CO <sub>2</sub> avoided converted to metric tons ÷ 0.039 tons per seedling grown for 10 years                                                                                |

The four load points (Load 100/75/50/25%) and their shares of the year come from Steps 3 and 4. The pump is assumed to be a single pump sized to meet and exceed the coil's design flow by the Pump Design Factor, as stated on the About screen.

## 9. Practicalities and Troubleshooting

### 9.1 Presenting and printing

The Results tab is the page to show. Print or export it once the inputs are settled – it carries the comparison, the savings, the financial summary and the rebate block on one sheet. To save a sheet as a PDF from the Pump Savings tab, use its PDF button (Section 4.4); on other tabs use Excel's File › Export or Print.

### 9.2 Language

Switching language on the Home tab re-labels the whole workbook, including the Results tab headings, the help messages and the chart legends. Numbers and units follow the separate Units drop-down. Your language choice is applied each time the workbook opens.

### 9.3 If something looks wrong

| Symptom                                                       | Check                                                                                                                                                                                                    |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A cell will not let you type.                                 | It is a calculated result. Compare it against the shading key in Section 2.4.                                                                                                                            |
| Savings look implausibly large.                               | Check the gap between Actual and Future Delta-T, and the Pump Design Factor on the Pump Savings tab.                                                                                                     |
| The financial results look wrong.                             | Almost always the investment figure. Confirm it holds the incremental cost of the Energy Valve over the valve that would have been fitted anyway.                                                        |
| Delta-T entries are rejected.                                 | The tool enforces a minimum gap of 2 degrees between Actual and Future, and a sensible range on Design. Adjust the pair together.                                                                        |
| A box says "The results compare six fixed Delta T scenarios". | You typed an Actual or Future Delta-T that is not one of the six scenarios (4, 6, 8, 10, 12, 14 °F). The tool adjusted it to the nearest one and the Home and Results tabs use that value (Section 3.1). |
| The operating-time total shows an error.                      | The shares for points A, B and C in Step 3 add up to more than 100%. Reduce one of them.                                                                                                                 |
| Numbers did not change after an edit.                         | Click away from the cell to commit the entry, or press F9 to recalculate.                                                                                                                                |
| Buttons and the About screen do nothing.                      | Macros are disabled. Click Enable Editing / Enable Content (Section 2.1).                                                                                                                                |
| Check for Updates reports a problem.                          | The update source could not be reached. The tool keeps working; try again later.                                                                                                                         |

This tool produces an estimate. Results depend on the accuracy of the inputs and on conditions specific to the installation. Questions, corrections and feature requests go to the maintainer named on the About screen.

## Appendix. Release History

| Release | Date       | Changes                                                                                                                                                                                                                                                                                                                                                                          |
|---------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.2     | 2026-09-24 | Added the standard admin keys used in every Belimo tool: F12 asks for the admin password and then shows and unprotects every sheet; F11 puts every sheet back exactly as it was. One password now protects every sheet in every tool. These keys are for maintainers and do not change how you use the tool.                                                                     |
| 4.1     | 2026-09-23 | Actual and Future Delta-T are now matched to the six scenarios the results use (an entry such as 7 or 13 is adjusted and you are told; an Actual below 4 no longer shows errors). The Min load point now counts its hours. The homes-equivalent figure uses the 12,258 kWh per home stated in its help text. Removed a stray period from the savings heading on the Results tab. |
| 4.0     | 2026-09-22 | Added automatic update checking through Belimo's update source. Added Check for Updates and Download Fresh Copy buttons to the About screen. Updates save through a normal Windows Save As dialog and never touch your current copy. The About screen's version line is now live. The About screen also has a Download User Manual button, which fetches this manual.            |
| Earlier | –          | Multi-language (English, Español, Português, Français) and US/Metric units; the Home tab reorganised around a Step 1 for language and units; the consolidated Results tab with the slider, greenhouse-gas equivalents, financial summary and Utility Rebate Support block.                                                                                                       |